SituatiONION V5.7

Statistical & Implementation Validation

Purpose

V5.7 tests the statistical and implementation stability
of the main V5 decodability findings.

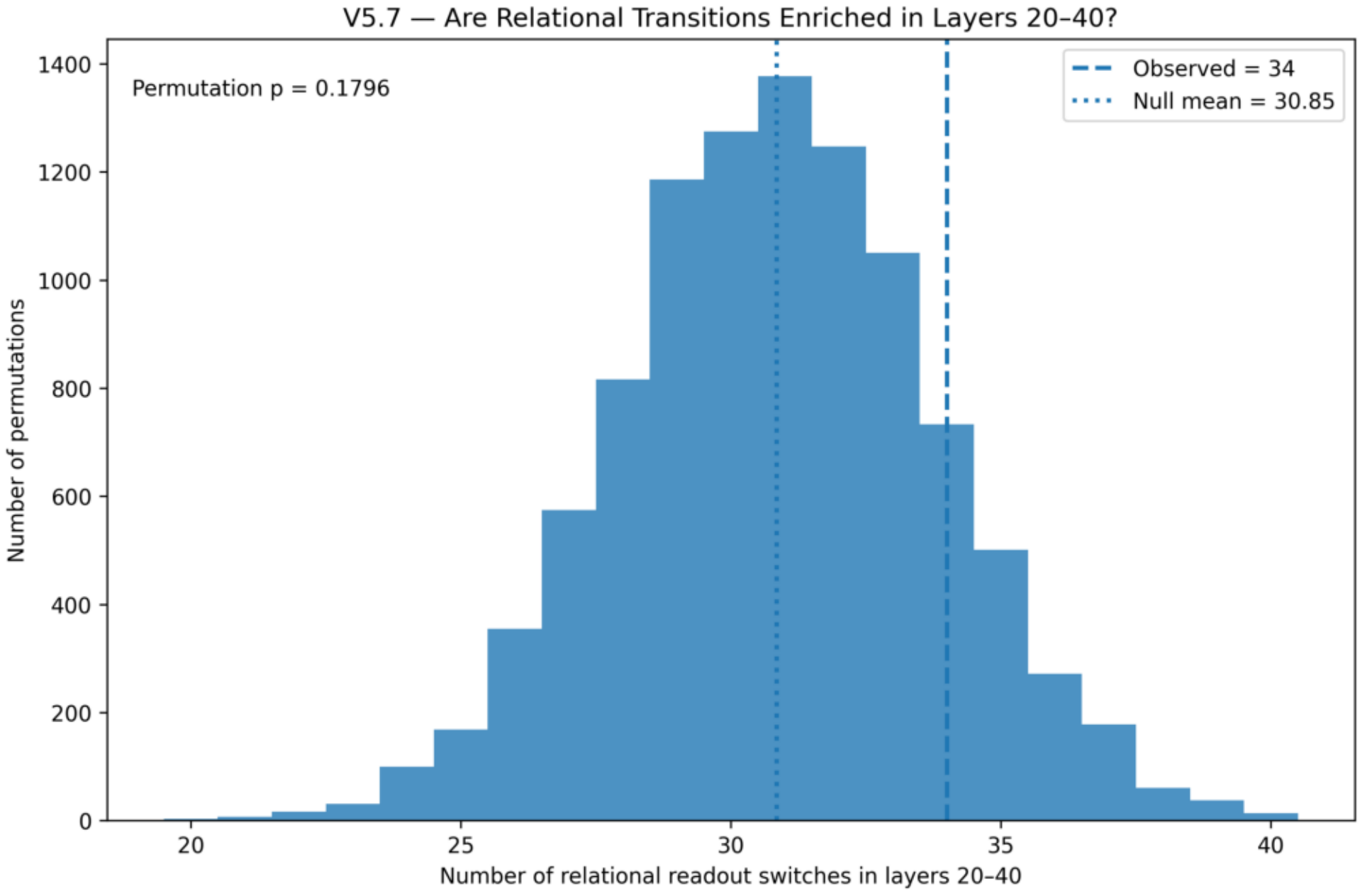
1. Transition Enrichment
   Layers 20–40: observed = 34, null mean = 30.85, p = 0.1796
   Layers 22–27: observed = 10, null mean = 8.77, p = 0.3512

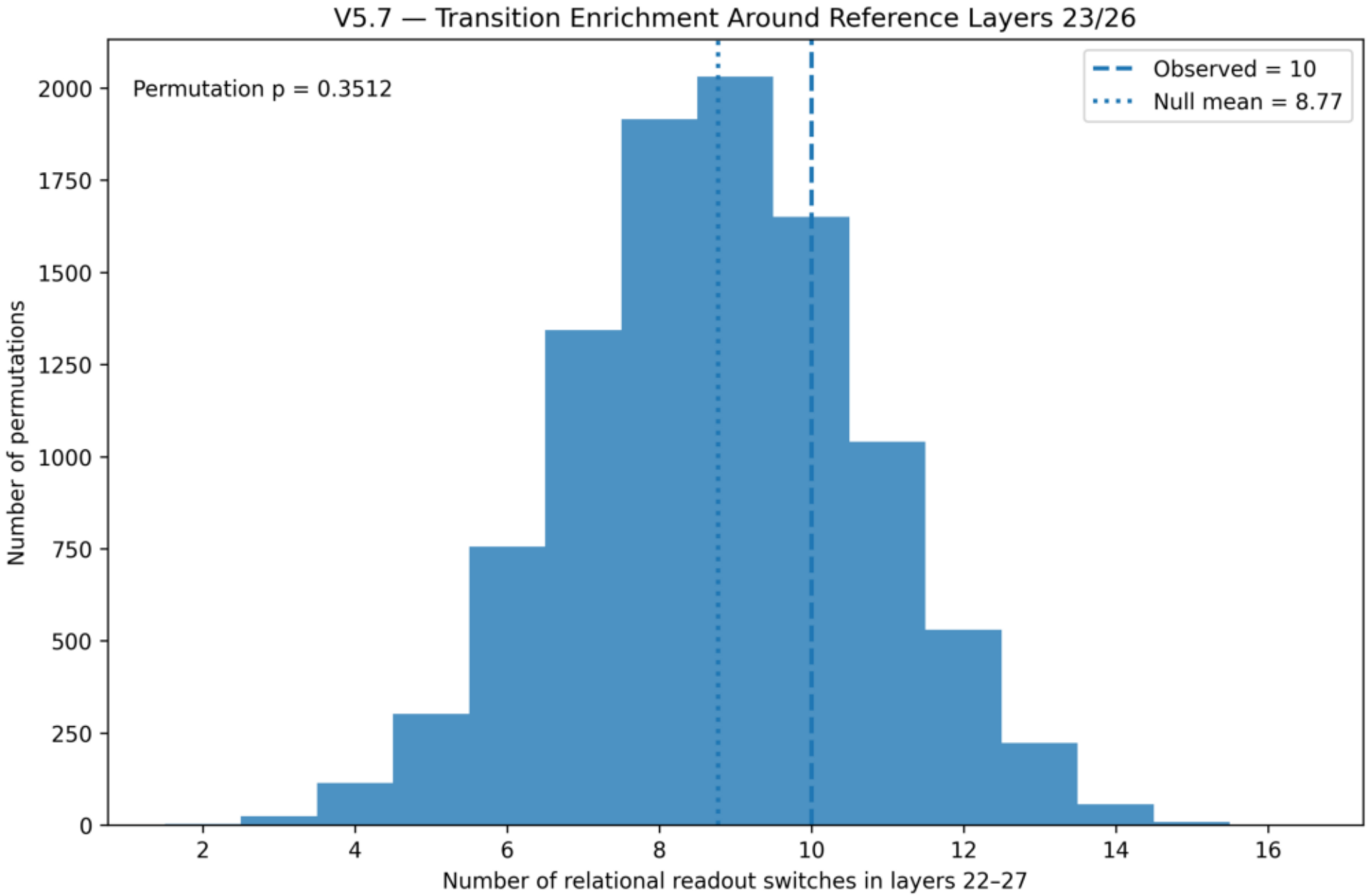
2. Paired Bootstrap of Conservative Advantage
   Bootstrap replicates = 5,000
   Full hidden-state probes are compared with their strongest nonchance controls.
   Temporal Relation: +0.500, 95% CI [0.500, 0.500]
   Cause Holder Role: +0.333, 95% CI [0.187, 0.500]
   Role Assignment: +0.250, 95% CI [0.083, 0.458]
   Event State: +0.000, 95% CI [0.000, 0.000]
   Polarity: +0.000, 95% CI [0.000, 0.000]

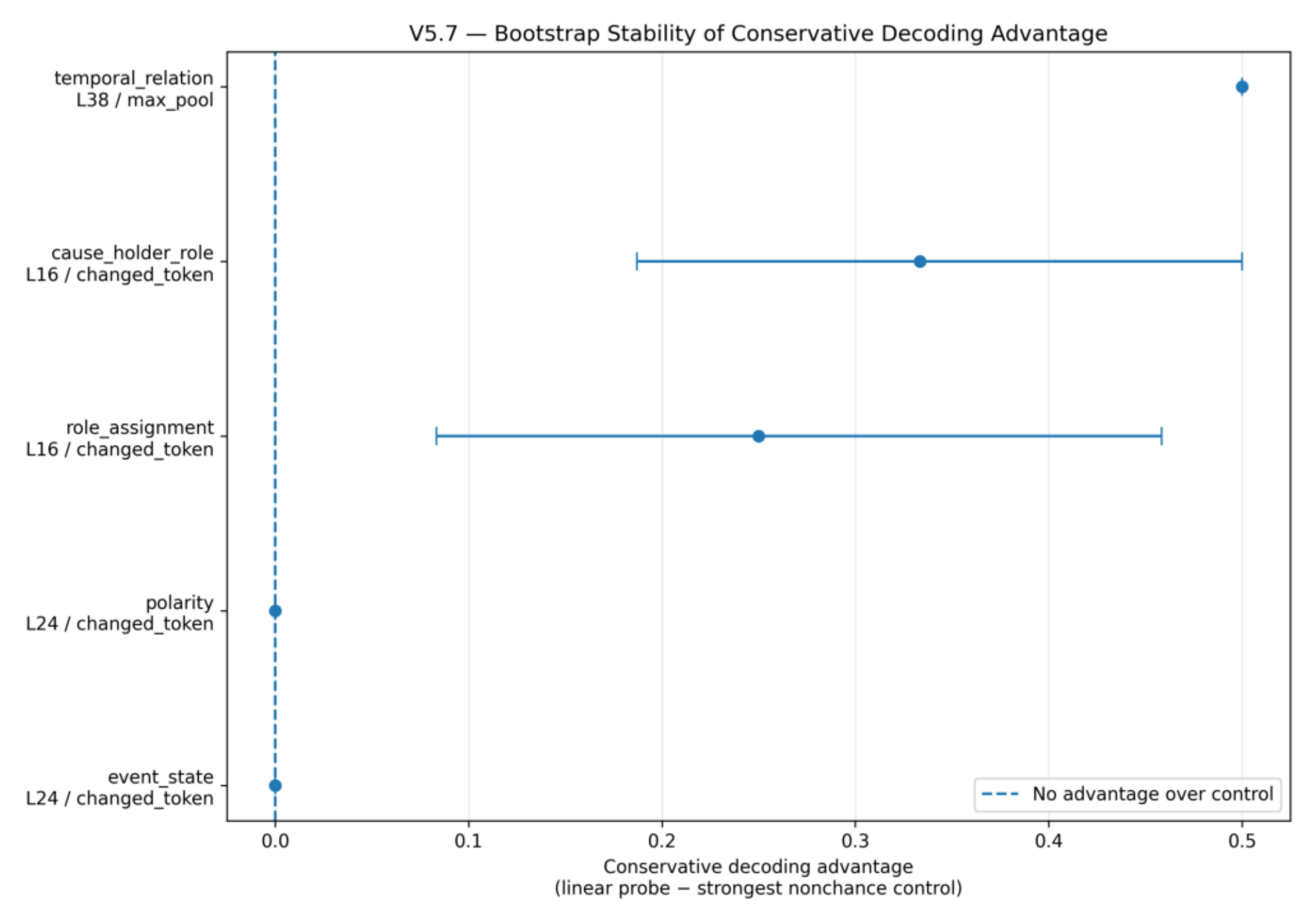
3. CPU/GPU Validation
   Pending independent CPU validation.

Interpretation

The selected relational configurations show positive control-adjusted decoding advantages.
Event state and polarity show no advantage over their strongest controls.
The permutation tests do not support a statistically localized transition in layers 20–40 or 22–27.
These analyses concern decodability and representational accessibility, not causal information flow.


V5.7 — Are Relational Transitions Enriched in Layers 20–40?
Permutation p = 0.1796
Observed = 34
Null mean = 30.85
Number of permutations
0
200
400
600
800
1000
1200
1400
20
25
30
35
40
Number of relational readout switches in layers 20–40


V5.7 — Transition Enrichment Around Reference Layers 23/26
Permutation p = 0.3512
Observed = 10
Null mean = 8.77
Number of permutations
0
250
500
750
1000
1250
1500
1750
2000
2
4
6
8
10
12
14
16
Number of relational readout switches in layers 22–27


V5.7 — Bootstrap Stability of Conservative Decoding Advantage
temporal_relation
L38 / max_pool
cause_holder_role
L16 / changed_token
role_assignment
L16 / changed_token
polarity
L24 / changed_token
event_state
L24 / changed_token
No advantage over control
0.0
0.1
0.2
0.3
0.4
0.5
Conservative decoding advantage
(linear probe − strongest nonchance control)